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$$\begin{aligned} & \int \frac{x+2}{\sqrt{3x-1}} dx \\ t = 3x-1 & \Rightarrow \frac{dt}{dx} = 3 \Rightarrow dx = \frac{dt}{3} \\ x = \frac{t+1}{3} & \Rightarrow x+2 = \frac{t+1}{3} + 2 = \frac{t+7}{3} \\ & = \int \frac{\frac{t+7}{3}}{\sqrt{t}} \cdot \frac{dt}{3} \\ & = \frac{1}{9} \int \frac{t+7}{\sqrt{t}} dt \\ & = \frac{1}{9} \int \left(t^{\frac{1}{2}} + 7t^{-\frac{1}{2}} \right) dt \\ & = \frac{1}{9} \left(\frac{2t^{\frac{3}{2}}}{3} + 14t^{\frac{1}{2}} \right) + c \\ & = \frac{2t^{\frac{3}{2}}}{27} + \frac{14t^{\frac{1}{2}}}{9} + c \\ & = \frac{2(3x-1)^{\frac{3}{2}}}{27} + \frac{14(3x-1)^{\frac{1}{2}}}{9} + c \\ & = \frac{2(3x-1)\sqrt{3x-1}}{27} + \frac{14\sqrt{3x-1}}{9} + c \end{aligned}$$

References